# Yongho Shin

Postdoctoral researcher
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## Research Interests

Online/approximation/learning-augmented algorithms for combinatorial optimization problems

## Education

**Ph.D. in Computer Science, Yonsei University, South Korea** Mar. 2018 – Aug. 2024

- Dissertation topic: Relaxing hard requirements of online optimization via learning augmentation and limited revocability
- Advisor: Hyung-Chan An

**B.S. in Computer Science, Yonsei University, South Korea** Mar. 2012 – Feb. 2018

- Awarded *high honors at graduation*

## Employment

**Institute of Computer Science, University of Wrocław, Poland** Nov. 2024 – present

- Postdoctoral researcher
- Advisor: Jarosław Byrka

## Research Papers

*Published Work*

[1] **Yongho Shin** and Phanu Vajanopath. Parsimonious learning-augmented online metric matching. *arXiv preprint arXiv:/2605.26886*, 2026. **To appear in ICML 2026.**

[2] Davin Choo, Billy Jin, and **Yongho Shin**. Learning-augmented online bipartite fractional matching. *Advances in Neural Information Processing Systems 38 (NeurIPS 2025)*, 7671–7686, 2025.

[3] Changyeol Lee, **Yongho Shin**, and Hyung-Chan An. Improved algorithms for overlapping and robust clustering of edge-colored hypergraphs: An LP-based combinatorial approach. *Advances in Neural Information Processing Systems 38 (NeurIPS 2025)*, 70823–70855, 2025.

[4] **Yongho Shin**, Changyeol Lee, Gukryeol Lee, and Hyung-Chan An. Improved learning-augmented algorithms for the multi-option ski rental problem via best-possible competitive analysis. *ACM Transactions on Algorithms 22(1)*, 11:1–31, 2025. *Conference version in ICML 2023 [5].*

[5] **Yongho Shin**, Changyeol Lee, Gukryeol Lee, and Hyung-Chan An. Improved learning-augmented algorithms for the multi-option ski rental problem via best-possible competitive analysis. In *Proceedings of the 40th International Conference on Machine Learning (ICML 2023)*, PMLR 202:31539-31561, 2023.

[6] Kangsan Kim, **Yongho Shin**, and Hyung-Chan An. Constant-factor approximation algorithms for parity-constrained facility location and $k$-center. *Algorithmica 85*, 1883–1911, 2023. *Conference version in ISAAC 2020 [8].*

[7] **Yongho Shin** and Hyung-Chan An. Making three out of two: Three-way online correlated selection. In *Proceedings of the 32nd International Symposium on Algorithms and Computation (ISAAC 2021)*, 49:1-49:17, 2021.

[8] Kangsan Kim, **Yongho Shin**, and Hyung-Chan An. Constant-factor approximation algorithms for the parity-constrained facility location problem. In *Proceedings of the 31st International Symposium on Algorithms and Computation (ISAAC 2020)*, 21:1-21:17, 2020.

*Preprints*

[9] Changyeol Lee, Dahoon Lee, Jongseo Lee, **Yongho Shin**, and Changki Yun. Optimal learning-augmented algorithm for online bidding. *arXiv preprint arXiv:2605.07349*, 2026.

[10] Fateme Abbasi, Martin Böhm, Jarosław Byrka, Matin Mohammadi, and **Yongho Shin**. Servicing matched client pairs with facilities. *arXiv preprint arXiv:2602.19680*, 2026.

[11] Fateme Abbasi, Hyung-Chan An, Jarosław Byrka, Changyeol Lee, and **Yongho Shin**. Chromatic correlation clustering via cluster LP. *arXiv preprint arXiv:2510.13446*, 2025.

[12] Jarosław Byrka and **Yongho Shin**. Online rounding for set cover under subset arrivals. *arXiv preprint arXiv:2507.13159*, 2025.

[13] **Yongho Shin**, Changyeol Lee, and Hyung-Chan An. On optimal consistency-robustness trade-off for learning-augmented multi-option ski rental. *arXiv preprint arXiv:2312.02547*, 2023. *Merged to [4].*

[14] **Yongho Shin**, Kangsan Kim, Seungmin Lee, and Hyung-Chan An. Online graph matching problem with a worst-case reassignment budget. *arXiv preprint arXiv:2003.05175*, 2020.

## Teaching

University of Wrocław, Poland

- ◇ Algorithms with predictions — Summer 2026

*Teaching assistant*, Yonsei University, South Korea

- ◇ CSI2103/CCO2103 Data structures — Spring 2018 – 2021, 2023, 2024
- ◇ CSI3108 Algorithm analysis — Fall 2018 – 2021, 2023
- ◇ AIC2130 Computer algorithms for AI applications — Fall 2023
- ◇ GEK6205 Design and analysis of optimization algorithms — Fall 2023

## Awards

*High honors at graduation*, Yonsei University — Feb. 2018

## Academic Service

- ◇ Reviewer, MFCS 2026
- ◇ Reviewer, IWOCA 2026
- ◇ Reviewer, ICML 2026
  - Gold reviewer (top 25%)
- ◇ Reviewer, Acta Informatica 2026
- ◇ Program committee member, EC 2026
- ◇ Reviewer, STOC 2026
- ◇ Reviewer, APPROX 2025

- ◇ Reviewer, ICALP 2025
- ◇ Reviewer, Mathematical Programming 2024
- ◇ Reviewer, WAOA 2024

## Talks and Presentations

*SNU Math Seminar*, SNU, Seoul, South Korea — Jan. 2026
- ◇ Title: Online rounding for set cover under subset arrivals

*Combinatorics Seminar*, Hanyang University, Seoul, South Korea — Jan. 2026
- ◇ Title: Primal-dual algorithms for edge-colored clustering

*SNU CSE Seminar*, SNU, Seoul, South Korea — Dec. 2025
- ◇ Title: Learning-augmented online bipartite fractional matching

*NeurIPS 2025*, San Diego, CA, USA — Dec. 2025
- ◇ Title: Improved algorithms for overlapping and robust clustering of edge-colored hypergraphs: An LP-based combinatorial approach
- ◇ Poster presentation

*NeurIPS 2025*, San Diego, CA, USA — Dec. 2025
- ◇ Title: Learning-augmented online bipartite fractional matching
- ◇ Poster presentation

*2025 Combinatorics Workshop*, IBS DIMAG, Daejeon, South Korea — Aug. 2025
- ◇ Title: Learning-augmented online bipartite fractional matching
- ◇ Contributed talk

*DGIST BRL AGSTA Seminar*, DGIST, Daegu, South Korea — Aug. 2025
- ◇ Title: Online rounding for set cover under subset arrivals

*Discrete Analysis Seminar*, Yonsei University, Seoul, South Korea — Aug. 2025
- ◇ Title: Online rounding for set cover under subset arrivals

*Combinatorial Optimization Seminar*, Yonsei University, Seoul, South Korea — Aug. 2025
- ◇ Title: Learning-augmented online bipartite fractional matching

*HALG 2025*, ETH Zurich, Zurich, Switzerland — June 2025
- ◇ Title: Learning-augmented algorithms for the multi-option ski rental problem
- ◇ Contributed talk and poster presentation

*COG Seminar*, University of Wrocław, Wrocław, Poland — May 2025
- ◇ Title: Learning-augmented online bipartite fractional matching

*COG Seminar*, University of Wrocław, Wrocław, Poland — Dec. 2024
- ◇ Title: Online correlated selection

*Discrete Analysis Seminar,* Yonsei University, Seoul, South Korea — June 2024
- ◇ Title: Three-way online correlated selection

*Discrete Math Seminar*, IBS DIMAG, Daejeon, South Korea — May 2024
- ◇ Title: Three-way online correlated selection

*ICML 2023*, Honolulu, HI, USA July 2023

- ◇ Title: Improved learning-augmented algorithms for the multi-option ski rental problem via best-possible competitive analysis
- ◇ Poster presentation

*Theory Tea*, Cornell University, Ithaca, NY, USA Dec. 2022

- ◇ Title: Three-way online correlated selection

*ISAAC 2021*, Fukuoka, Japan (virtual) Dec. 2021

- ◇ Title: Making three out of two: Three-way online correlated selection

*AAAC 2021*, Tainan, Taiwan (virtual) Oct. 2021

- ◇ Title: Making three out of two: Three-way online correlated selection

*ISAAC 2020*, Hong Kong, China (virtual) Dec. 2020

- ◇ Title: Constant-factor approximation algorithms for the parity-constrained facility location problem

## Experience

*Co-organizer* of Yonsei CS theory student group, Yonsei University, South Korea Jan. 2023 – Feb. 2024

- ◇ Initiated a reading group of TCS students in and out of Yonsei University
- ◇ Organized seminar talks on various topics including mechanism design and quantum computing

*Intern*, Cornell University, USA Sept. 2022 – Dec. 2022

- ◇ Director: David B. Shmoys

*Undergraduate intern*, Yonsei University, South Korea Jan. 2017 – Feb. 2018

- ◇ Advisor: Hyung-Chan An

*Undergraduate voluntary tutor*, Yonsei University, South Korea

- ◇ CSI3108 Algorithm analysis Fall 2016, 2017
- ◇ CSI2103 Data structures Spring 2017

*Web programmer*, Republic of Korea Air Force, South Korea Nov. 2013 – Aug. 2015

- ◇ In fulfillment of mandatory military service